Chaitya Shah

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Education

Columbia University - Fu Foundation School of Engineering and Applied Science — New York, NY
Master of Science in Data Science | GPA: 3.79 /4 — Expected Dec 2026

University of Mumbai - Dwarkadas J. Sanghvi College of Engineering — Mumbai, India
Bachelor of Technology in Computer Engineering | GPA: 8.76/10 — Dec 2021 - May 2025

Experience

JPMorganChase — New York, NY
Machine Learning Center of Excellence, Summer Associate — Jun 2026 - Aug 2026

- Post-trained LLM's via SFT (QLoRA) to build a unified framework to extract structured trading information from **1.5M+ daily conversations** across 9 financial asset classes, achieving **96–107%** of asset-specific model performance.
- Demonstrated **cross-asset knowledge transfer**, achieving **82% event detection** on domains without training annotations.
- **Benchmarked open-source LLM scaling**, quantifying a 3.4-point F1 gain from 8B to 24B models at 4× GPU requirements.
- Explored **Dynamic LoRA** routing/merging, enabling domain-specific updates, cutting compute 80% over full retraining.

SimPPL — New York, NY
Founding AI Engineer — Aug 2023 - May 2025

- Spearheaded a team of 4 to launch an AI-powered **disinformation detection** tool for 9 social media platforms, with grants from **Mozilla and Google (exploreCSR)**, helping New York Public Radio boost engagement for 20K+ users.
- Deployed a **full-stack production system** with a Next.js, Flask API's, PostgreSQL and Redis data layers, containerized with Docker and served via Nginx on **AWS**, **processing 1M+ user interactions** to drive real-time content recommendations.
- Engineered an **AI Agent** using LangGraph to analyze **120k+ social media posts** for our **UNICEF-incubated tech venture**.
- Enabled journalists in **Bangladesh** to identify accounts responsible for coordinated harassment of minorities; investigations using the platform contributed to **Meta removing accounts with 100M+ views**.

Avrio Energy — Berlin, Germany
Data Science Intern — Jun 2024 - Jul 2024

- Optimized energy usage across **70+ McDonald's outlets** in 22 countries by engineering a data pipeline to track appliance level consumption and visualize real-time insights via Power BI, **reducing operational costs by 15%**.
- Automated real-time fault detection with Airflow alerts and a **92% accurate XgBoost model**, cutting energy waste by 21%.

Indian Institute of Technology Patna — Patna, India
Research Intern — May 2023 - Nov 2023

- Designed a fraud detection system on **250k+ Ethereum transactions** using Graph Neural Networks (Temporal GATs) and Directed Random Walks to classify fraudulent nodes based on behavioral patterns, achieving **85% accuracy**.

Infiheal — Mumbai, India
Artificial Intelligence Intern — Jun 2023 - Sep 2023

- Engineered a **RAG-based mental health chatbot** with LangChain & Haystack, **fine-tuned LLaMA** on 500k+ user chat transcripts to handle sensitive user queries, reducing hallucinations and **increasing user retention by 12%**.
- Deployed scalable ML architecture on AWS with Terraform & Docker, reducing manual setup time by **5+ hrs/week**.

Publications

- Shah, C., et al. "Can Social Media Platforms Transcend Political Labels? An Analysis of Neutral Conversations on Truth Social". International **AAAI** Conference on Web and Social Media 2024. **https://doi.org/10.36190/2024.17**

Projects

Automatic Redaction Extraction Pipeline | Research Assistant - Prof. Vishal Misra

- Developing a cost-efficient **VLM pipeline** to identify and classify redactions across 3,000+ declassified CIA intelligence documents, achieving 91% precision and enabling **multimodal prediction** of the underlying redacted information.

Dynamic KV Cache Compression for Reasoning in LLMs

- Applied Compressed Context Memory for chain-of-thought reasoning by training a **token-level classifier** with LoRA adapted LLaMA-3.1-8B, using GPT-4o-mini **knowledge distillation** to dynamically compress GPU KV caches.
- Achieved **1.60× memory compression** with improved reasoning accuracy 26% → 32% and <0.4% inference overhead.

Skills

Programming & ML: Python, C++, SQL, MATLAB, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, OpenCV, NLTK
LLMs and GenAI: SFT, DPO, LoRA/PEFT, RAG, CUDA, LLM Evaluation, Knowledge Distillation, RLHF, LangChain, LangGraph
Systems & Infrastructure: AWS, GCP, Docker, Linux, PostgreSQL, Redis, BigQuery, FastAPI, Nginx